

Detailed Performance Summary For R-

513a_Qx_100to20_dTe_const_Tc_41to20C

Project: 30XW_China

Prepared By: Mikhail Terekhov

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AquaForce Greenspeed Water Cooled Screw Chiller



Unit Information

Tag Name: **R-513a_Qx_100to20_dTe_const_Tc_41to20C**
Model Number: **30XW-VC459**
Condenser Type: **Water Cooled**
Nameplate Voltage: **400-3-50** V-Ph-Hz
Quantity: **3**
Manufacturing Source: **Shanghai, China**
Chiller Type: **Cooling**
Operating Mode: **Cooling**
Refrigerant: **R-513A**
Minimum Capacity: **20.00** %
Shipping Weight: **9421** kg
Operating Weight: **9327** kg
Unit Length: **4524** mm
Unit Width: **1745** mm
Unit Height: **2632** mm

Performance Information

Cooling Capacity: **1 462** kW
Heating Capacity: **1 791** kW
Total Unit Power: **363.1** kW
Cooling Efficiency (COPR): **4.026** kW/kW
Heating Efficiency (COPH): **4.932** kW/kW

Acoustics Information

A-Weighted Sound Pressure Level: **81** dBA

Evaporator Information

Fluid Type: **Fresh Water**
Fouling Factor: **0.0180** (sqm-K)/kW
Leaving Temperature: **6.00** °C
Entering Temperature: **13.00** °C
Fluid Flow: **49.78** L/s
Pressure Drop: **28.5** kPa

Condenser Information

Fluid Type: **Propylene Glycol**
Brine Concentration: **45.0** %
Fouling Factor: **0.0440** (sqm-K)/kW
Leaving Temperature: **49.00** °C
Entering Temperature: **41.00** °C
Fluid Flow: **58.89** L/s
Pressure Drop: **42.7** kPa

Electrical Information

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Unit Voltage:.....**400-3-50** V-Ph-Hz
Standby Power:.....**0.05000** kW
Minimum Voltage:.....**360** Volts
Maximum Voltage:.....**440** Volts
Power Factor:.....**0.93**

Amps (Un)	Electrical Circuit 1	Electrical Circuit 2
Max Unit Current Draw (RLA)	567.0	---
Nominal Start Up Current (ICF)	384.0	---
Nominal Unit Current Draw (A)	384.0	---

Accessories and Installed Options

Opt. 282 EMC C2 Classification

Opt. 001A Integrated VFD

Opt. 150E Cooling Chiller Cond Max Leaving Temp 50C

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Load Line

Unit Performance	**	***								(1)
User-specified Cooling Capacity, kW		1000.0	900.0	800.0	700.0	600.0	500.0	400.0	300.0	200.0
Percent of User-specified Cooling Capacity, %	146	100.00	90.00	80.00	70.00	60.00	50.00	40.00	30.00	20.00
Percent Full Load Cooling Capacity, %	100.0	68.41	61.57	54.73	47.89	41.05	34.21	27.37	20.52	13.68
Percent of Full Load Power, %	100.0	71.75	64.76	55.80	41.15	31.94	22.95	16.71	10.55	6.812
Unloading Sequence	Default	Default	Default	Default	Default	Default	Default	Default	Default	Default
Cooling Capacity, kW	1 462	1 000.	900.0	800.0	700.0	600.0	500.0	400.0	300.0	200.0
Heating Capacity, kW	1 791	1 236	1 110.	979.0	831.4	701.6	572.9	452.8	333.1	220.8
Total Unit Power, kW	363.1	260.5	235.1	202.6	149.4	116.0	83.34	60.68	38.31	24.74
Nominal Unit Current Draw (RLA)	554.60	397.6	360.3	313.1	233.6	181.8	130.6	95.1	60.0	---
Cooling Efficiency kW/kW	4.026	3.838	3.828	3.949	4.685	5.173	5.999	6.592	7.830	8.085
Heating Efficiency, kW/kW	4.932	4.745	4.723	4.832	5.565	6.049	6.874	7.463	8.695	8.925
A-Weighted Sound Pressure Level, dBA	81	77	77	78	74	73	73	73	73	-
Evaporator Data										
Fluid Entering Temperature, °C	13.00	13.00	13.00	13.00	13.00	12.86	11.72	10.57	9.43	8.29
Fluid Leaving Temperature, °C	6.00	6.00	6.00	6.00	6.00	6.00	6.00	6.00	6.00	6.00
Fluid Flow Rate, L/s	49.78	33.99	30.59	27.19	23.79	20.81	20.80	20.80	20.80	20.80
Pressure Drop, kPa	28.5	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0
Fouling Factor, (sqm-K)/kW	0.0180	0.0180	0.0180	0.0180	0.0180	0.0180	0.0180	0.0180	0.0180	0.0180
Condenser Data										
Fluid Entering Temperature, °C	41.00	41.00	40.00	39.00	35.00	32.00	29.00	26.00	23.00	20.00
Fluid Leaving Temperature, °C	49.00	49.00	47.24	45.39	40.43	36.59	32.76	28.97	25.19	21.45
Fluid Flow Rate, L/s	58.89	40.52	40.48	40.45	40.38	40.32	40.27	40.21	40.17	40.10
Pressure Drop, kPa	42.7	21.5	21.7	21.9	22.6	23.1	23.7	24.4	25.1	25.9
Fouling Factor, (sqm-K)/kW	0.0440	0.0440	0.0440	0.0440	0.0440	0.0440	0.0440	0.0440	0.0440	0.0440

Note:

(**): Column displaying effective unit full load performances

(***): Column displays unit performance at the 'required capacity' and design conditions specified

(1) Compressor cycling is expected. Stated performances are average.

Note: Performance is at nominal voltage. Chiller may unload if job site voltage is lower than 400 volts.

Sound pressure level was determined in accordance with AHRI Standard 575. Sound pressure level is the free field sound data (the unit is placed in free field over a reflecting plane).

Outside the scope of AHRI Water-Cooled Water-Chilling and Heat Pump Water-Heating Packages Certification Program or not optionally certified, but is rated in accordance with AHRI Standard 550/590 (I-P) and AHRI Standard 551/591 (SI).

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